



Exploring Neuroimaging Association Scores in adulthood ADHD and middle-age trajectories

Cibele Edom Bandeira^{a,b}, Eugenio Horacio Grevet^{a,c,d,e}, Eduardo Schneider Vitola^{a,b},
Bruna Santos da Silva^{a,f}, Renata Basso Cupertino^g, Felipe Almeida Picon^a, Lucas Toshio Ito^{h,i},
Maria Eduarda de Araujo Tavares^{a,c,j}, Diego Luiz Rovaris^{a,b}, Oliver Grimm^k, Claiton Henrique
Dotto Bau^{a,c,e,j,l,*}

^a ADHD Outpatient Program, Clinical Research Center, Department of Psychiatry, Hospital de Clínicas de Porto Alegre, Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil

^b Laboratory of Physiological Genomics of Mental Health (PhysioGen Lab), Instituto de Ciencias Biomedicas da Universidade de São Paulo, São Paulo, Brazil

^c Laboratory of Developmental Psychiatry, Center of Experimental Research, Hospital de Clínicas de Porto Alegre, Porto Alegre, Brazil

^d Department of Psychiatry, Faculty of Medicine, Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil

^e Graduate Program in Psychiatry and Behavioral Sciences, Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil

^f Department of Basic Health Sciences, Federal University of Health Sciences of Porto Alegre (UFCSA), Porto Alegre, Brazil

^g Department of Psychiatry, University of California San Diego, San Diego, CA, USA

^h Department of Biochemistry, Universidade Federal de São Paulo, São Paulo, Brazil

ⁱ Laboratory of Integrative Neuroscience, Universidade Federal de São Paulo, São Paulo, Brazil

^j Graduate Program in Genetics and Molecular Biology, Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil

^k Department of Psychiatry, Psychosomatic Medicine and Psychotherapy, University Hospital, Goethe University, Frankfurt, Germany

^l Department of Genetics, Institute of Biosciences, Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil

ARTICLE INFO

Keywords:

ADHD
Midlife clinical trajectories
Adults
Neuroimaging association scores

ABSTRACT

Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder associated with brain differences in children, but not in adults. A combined evaluation of the regional brain differences could improve statistical power and, consequently, allow the detection of possible effects in adults. Thus, our aim is to verify whether Neuroimaging Association Scores (NAS) are associated with adulthood ADHD and clinical trajectories of the disorder in midlife. Clinical and neuroimaging data were collected for 121 subjects with ADHD (mean age: 47.1 ± 10.5 ; 43% male) and 82 controls (mean age: 38.2 ± 9.0 ; 54.9% male). Cases were assessed seven and thirteen years after baseline diagnosis, and their clinical trajectories were classified as stable if they fulfilled ADHD diagnosis in all assessments or unstable if they presented remission and recurrence of symptoms. Neuroimaging data were acquired in the last clinical assessment (thirteen years after baseline) and NAS were calculated as a weighted sum of the associations previously reported by meta-analyses for three types of structural brain modalities: cortical thickness, cortical surface area, and subcortical volume. The NAS for cortical surface area was higher in cases compared to controls. No association was found for NAS and number of symptoms of ADHD or clinical trajectories. The fact that differences were restricted to ADHD diagnostic status suggests a susceptibility effect that is not extended to subtle aspects of the disorder. Our results also suggest that evaluating overall effects may have advantages especially when applied to adult ADHD samples.

1. Introduction

Attention-Deficit/Hyperactivity Disorder is a neurodevelopmental disorder characterized by inappropriate levels of inattention and/or

hyperactivity/impulsivity, first manifested in childhood and often persisting in adult life (Faraone et al., 2006). In neuroimaging studies, ADHD has been related to differences in structural brain aspects (Firouzabadi et al., 2022). Mega-analyses conducted by ENIGMA (Enhancing

* Corresponding author. Departamento de Genética, Instituto de Bociências, Universidade Federal do Rio Grande do Sul, UFRGS, Avenida Bento Gonçalves, 9500, Porto Alegre, RS, CEP: 91501-970, Brazil.

E-mail address: claiton.bau@ufrgs.br (C.H.D. Bau).

<https://doi.org/10.1016/j.jpsychires.2024.06.025>

Received 17 December 2023; Received in revised form 29 May 2024; Accepted 17 June 2024

Available online 17 June 2024

0022-3956/© 2024 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Neuro Imaging Genetics through Meta Analysis) consortium reported that children with ADHD presented smaller subcortical volumes of accumbens, amygdala, caudate, hippocampus, putamen and intracranial volume. Adolescents also presented smaller hippocampus volume (Hoogman et al., 2017).

For cortical measures, children with ADHD presented smaller surface area in 24 out of 34 regions evaluated, including total surface area. Additionally, the fusiform, parahippocampal, and precentral gyri and the temporal pole were thinner in ADHD (Hoogman et al., 2019). Among adults, who were only half the sample size of the children's group, there were no significant differences, which the authors hypothesized could be attributed to low statistical power or to an age-related attenuation of the effects.

Deep learning neural network classification models applied to the ENIGMA database revealed that these models can differentiate ADHD from controls, even in adults (Zhang-James et al., 2021). Yet, as expected, prediction performance was better in the child sample. These findings are in line with studies reporting cortical differences in adults with ADHD in specific regions (dorsal attentional network and limbic cortices; Proal et al., 2011) and in total volume of gray matter (Klein et al., 2021).

From a longitudinal perspective, ADHD has been associated with developmental delays in both cortical (Shaw et al., 2007, 2012) and subcortical (Wang et al., 2021) regions, as well as a slightly higher growth rate in the anterior corpus callosum compared to controls, from childhood through adolescence (Gilliam et al., 2011). Furthermore, young adults with a history of childhood and persistent ADHD exhibited decreased cortical surface area in specific brain regions, such as the bilateral parahippocampal gyri, left paracentral gyrus, and right transverse temporal gyrus, compared to those who experienced symptom remission (Luo et al., 2020). Considering follow-up trajectories since childhood, ADHD persists at middle age presented nominally thinner cortex in the insula, parahippocampus, medial occipital cortex, and prefrontal regions compared to remitters (Proal et al., 2011).

Taken together, these results demonstrate the dynamic nature of structural brain differences in ADHD. Multifactorial disorders such as ADHD, however, often exhibit neuroimaging association with small effect sizes, making it difficult to detect robust, replicable differences through conventional means. To overcome this challenge, the combined evaluation of regional effects has been proposed as a more effective methodology, increasing the analysis's statistical power. This approach has given rise to the development of neuroimaging liability scores (Kochunov et al., 2019; Axelrud et al., 2021; Lancaster et al., 2022; Rodrigue et al., 2023), which calculate a weighted sum of structural MRI alterations associated with specific phenotypes. Notably, this score predicted treatment resistance in schizophrenia better than any individual tract-specific white matter measure (Kochunov et al., 2019).

Neuroimaging liability scores were previously tested in relation to ADHD in a Brazilian sample (Axelrud et al., 2021). ADHD Neuroimaging Association Scores (NAS) for surface area were associated with baseline and follow-up general psychopathology, externalizing symptoms, and cognitive measures in a sample of children and adolescents (Axelrud et al., 2021). Our goal is to use NAS to better understand how brain differences might be linked to ADHD and its clinical trajectories among adults. We hypothesize that cases would present higher NAS compared to controls. We expected that the same pattern would be observed for related phenotypes, with a stable trajectory and higher number of ADHD symptoms associated with higher NAS.

2. Material and methods

2.1. Sample

All participants included in the sample were fully informed of study procedures in all time points (first, second, and third assessments) and provided signed informed consent, approved by the institutional review

board of HCPA (IRB 0000921). This work was carried out following the Declaration of Helsinki.

The sample was composed of 121 cases and 82 controls. Cases were first assessed in the Adult Division of the ADHD Outpatient Program at the Hospital de Clínicas de Porto Alegre (HCPA). Diagnosis was conducted using the Portuguese version of the Kiddie Schedule for Affective Disorders and Schizophrenia (K-SADS-E) (Mercadante et al., 1995) adapted for adults (Grevet et al., 2005). The presence of psychiatric comorbidities was evaluated through the Structured Clinical Interview for DSM-IV (SCID-I) (First et al., 1998). At baseline, clinical evaluations on the 121 cases revealed that 60 (49.6%) had the combined presentation, 54 (44.6%) manifested inattention presentation and 7 (5.8%) had the hyperactivity/impulsivity presentation of the disorder.

Follow-up data were collected through two clinical evaluations seven and thirteen years after baseline diagnosis. Considering all three clinical assessments, trajectories were defined as stable/persistent when subjects fulfilled ADHD diagnosis in all assessments and unstable/fluctuating when subjects presented remission and recurrence of symptoms along the evaluations. For a more detailed description of the baseline and follow-up clinical assessments, see Grevet et al. (2005) and Grevet et al. (2024), respectively.

Controls were blood donors with a negative screening for ADHD, assessed at the same hospital (HCPA). All participants fulfilled the following inclusion criteria: (i) being Brazilian of predominantly European descent and (ii) aged at least 18 years old. Exclusion criteria were (i) the presence of significant neurological disease (e.g., delirium, dementia, epilepsy, and/or head trauma), (ii) history of psychotic symptoms, and (iii) estimated intelligence quotient (IQ) ≤ 70 .

2.2. Neuroimaging data and NAS calculation

Structural T1-weighted MRI data were acquired in Siemens Magnetom Spectra 3T scanner with a 16-channel phased-array head coil. MRI data were collected in the third clinical assessment, where a questionnaire to evaluate handedness was included in the clinical follow-up protocol (Edinburgh Handedness Inventory) (Oldfield 1971). For a detailed description of the MRI data acquisition, preprocessing, and quality control in cases and in controls, see (Bandeira et al., 2021). Briefly, T1-weighted 3D magnetization prepared rapid acquisition with gradient echo (MPRAGE) sequence acquired had 192 slices, flip angle = 7° , TE = 2.55 ms, TR = 2530 ms, TI = 1100 ms, matrix size = 256×256 , and slice thickness = 1 mm, lasting 6:02 min. Anatomical segmentation was performed with the automated pipeline from Freesurfer version 5.3 (Fischl 2012) and quality control procedures followed ENIGMA protocol (Hibar et al., 2015).

The NAS were calculated for ADHD following the code available online (<https://gitlab.com/sanps/nas>) provided by Axelrud et al. (2021). The scores were calculated separately for three structural brain modalities: cortical thickness, cortical surface area, and subcortical volume. Within each modality, only regions that had already been tested for ADHD and had results described in the latest ENIGMA mega-analyses were included. In Supplementary Tables 1–6, the regions included are specified (34 for cortical thickness, 34 for surface area, and 7 for subcortical volumes), along with the P-values and effect sizes (Cohen's D) for its association with ADHD from (Hoogman et al., 2017, 2019) studies.

Since our sample is composed of adults, the scores were calculated using the ENIGMA ADHD Cohen's D reported exclusively for adults' sample (from now on called "adults-NAS"). To verify if statistics derived from a larger sample could be better to calculate NAS and to possibly detect subtle neuroimaging differences in our sample of adults, we also calculated the scores using the ENIGMA ADHD Cohen's D estimated in the whole sample (children + adolescents + adults, from now on called "all-NAS"). At first, the scores were calculated at P-thresholds (Pt) of 0.05, 0.5, and 1, considering P-values reported by ENIGMA. One additional score (referred as Pt_{FDR}) was calculated including only the regions

significantly associated with ADHD in ENIGMA mega-analyses after false discovery rate correction. The number of regions included in each score varied according to the significance threshold used (similar to the Pt strategy in the polygenic risk score calculation).

The scores were calculated in our sample using Rstudio version 4.3.2 (RStudio, Boston, USA). First, each structural measure for the regions included was regressed out for covariates (age, sex, and any other possible confounder) through multivariate regressions. Next, the residuals were transformed into z scores considering the mean and standard deviation from the complete sample (cases and controls). Then, each standardized measure was multiplied by the effect size (Cohen's D) reported in ENIGMA ADHD mega-analyses. Finally, the products of these multiplications were summed to generate the scores for each modality (cortical thickness, surface area, and subcortical volume). These scores reflect the similarity to an ADHD brain, as assessed in the discovery samples (Hoogman et al., 2017, 2019).

To assess internal consistency, Cronbach's alpha reliability indexes were generated for each score using the psych R package v.2.2.5 (Rev-elle 2017). A Cronbach's alpha higher or equal to 0.7 was considered acceptable (George and Mallery 2003), and this information was used to guide the selection of P-thresholds with the best reliability for subsequent statistical analyses. The Cronbach's alpha values for all scores, along with the number of regions included at each P-threshold, are presented in Supplementary Table 7. Ultimately, only NAS with Pt of 1 were included in the analyses since this was the threshold with the highest Cronbach's alpha in almost all modalities.

2.3. Statistical analyses

The association between NAS and ADHD diagnosis was evaluated through Student's *t*-test (after confirming the normality of data distribution). Subsequent analyses focused solely on cases, where the relationship between NAS and the number of ADHD symptoms at baseline was evaluated through Pearson's correlation test. Finally, the relation between NAS and ADHD clinical trajectories (stable vs unstable) was assessed with Student's *t*-test. Sensitivity analysis was conducted to investigate if associations remained significant after adjusting for possible confounders.

Specifically for trajectory analyses, nine cases were excluded. Among these, three remitted ADHD diagnoses in the second and third assessments and were excluded since they composed a trajectory with low frequency. Six of them were using stimulants and referred symptoms attenuation or remission, being excluded once the improvement reported could be attributable to medication effect. From the remaining 112 cases, 88 coursed a stable persistent trajectory, and 24 presented an unstable trajectory.

Bonferroni multiple testing correction was applied considering the outcomes (ADHD diagnosis, number of symptoms, and trajectories) but not the NAS modalities (cortical surface area, cortical thickness and subcortical volumes) since they are significantly correlated and therefore not independent. Thus, results were considered significant if they presented a $P < 0.017$ ($0.05/3$ outcomes).

3. Results

Sample characteristics are shown in Table 1. Comparisons between the frequency of Bipolar Disorder and Substance Use Disorder in cases and controls was not possible due to the low number of representants in the control group. Handedness, assessed by the Edinburgh inventory, showed that 112 cases (93.3%) and 69 controls (84.1%) were right-handed ($\chi^2 = 4.414$; $P = 0.036$).

Overall, the NAS presented better Cronbach's alphas at higher P-thresholds. Due to the few significant findings for ADHD and neuroimaging in adults in the literature, it was possible to calculate only adult-NAS with broader Pt when Cohen's D derived from adults' discovery sample was used. From those, only NAS for cortical surface area

Table 1
Participants characteristics.

	ADHD diagnosis			Clinical trajectories		
	Cases (n = 121)	Controls (n = 82)	P	Stable (n = 88)	Unstable (n = 24)	P
Age (years)	47.1 (10.5)	38.2 (9.0)	1.1E-09	47.2 (10.3)	44.3 (10.6)	0.224
Sex (male)	52 (43.0)	42 (54.9)	0.096	32 (36.4)	16 (66.7)	0.008
Lifetime comorbidities ^b						
Major Depressive Disorder	59 (48.8)	36 (43.9)	0.496	46 (52.3)	10 (41.7)	0.357
Bipolar Disorder ^a	33 (27.3)	3 (3.7)	-	26 (29.5)	6 (25.0)	0.662
Generalized Anxiety Disorder ^a	23 (19.0)	13 (16.0)	0.590	19 (21.6)	4 (16.7)	-
Substance Use Disorder ^a	20 (16.5)	4 (4.9)	-	15 (17.0)	4 (16.7)	-

Data was expressed as mean (SD), or n (%).
^a Considering the low frequency of Bipolar Disorder, Generalized Anxiety Disorder, and Substance Use Disorder in controls or in the unstable trajectory (n < 5), individual comparisons were not possible.
^b The lifetime combined presence of any comorbidity included in the table slightly differed between the groups ($P < 0.2$), and therefore this composite variable was included in the sensitivity analyses as a covariate.

presented acceptable Cronbach's alpha (≥ 0.7). Using the Cohen's D reported for children + adolescents + adults, all scores presented acceptable Cronbach's alpha except for the Pt of 0.05 and P_{FDR} in cortical thickness. These values of Cronbach's alpha were expected since the discovery sample reported more associations in the whole sample than in adults.

The results for association analyses between NAS and ADHD diagnosis and number of symptoms and between NAS and midlife clinical trajectories are shown in Table 2. For NAS calculated with Cohen's D estimated in the adult sample (adult-NAS), cortical surface area scores were higher in cases compared to controls ($P = 0.008$; Fig. 1A). The same pattern was observed for NAS calculated with Cohen's D from all ages (all-NAS; $P = 0.004$; Fig. 1B). We further test if these scores (all- and adults-NAS for cortical surface area) also differ between ADHD presentations, with no significant associations. Within cases, NAS were not significantly associated with the number of symptoms at baseline or with clinical trajectories (Table 2). All-NAS for subcortical volumes, however, was nominally higher in unstable compared to stable trajectory ($P = 0.045$; Table 2).

For the sensitivity analyses, the presence of any lifetime comorbidity, considering Major Depressive Disorder, Bipolar Disorder, Generalized Anxiety Disorder, and Substance Use Disorder, and the continuous score of laterality provided by the Edinburgh inventory were included as covariates. Cronbach's alpha values for all scores calculated considering these covariates, and the number of regions included at each P-threshold are presented in Supplementary Table 8. Results remained similar, except for the lack of nominal association between subcortical volumes all-NAS and ADHD clinical trajectories (Supplementary Table 9).

4. Discussion

The significant associations between NAS for cortical surface area and ADHD diagnosis in adulthood provides evidence for the advantages of this method, particularly in its ability to capture overall neurobiological effects and its increased statistical power. It is noteworthy that previous neuroimaging studies found more pronounced effects in children than in adults (Hoogman et al., 2017, 2019; Zhang-James et al., 2021). By joining different regional effects, we were able to detect structural brain differences in adults with ADHD, suggesting that NAS

Table 2
Association between NAS, ADHD diagnosis, number of symptoms and clinical trajectories.

	ADHD diagnosis			ADHD symptoms		Clinical trajectories		
	Cases (n = 121)	Controls (n = 82)	P	r	P	Stable (n = 88)	Unstable (n = 24)	P
<i>ADHD adults-NAS</i>								
Cortical surface area	0.210 (0.11)	−0.311 (0.17)	0.008	0.146	0.111	0.189 (0.13)	0.398 (0.31)	0.473
<i>ADHD all-NAS</i>								
Cortical thickness	−0.101 (0.08)	0.149 (0.10)	0.057	−0.016	0.861	−0.065 (0.10)	−0.353 (0.15)	0.169
Cortical surface area	0.677 (0.34)	−0.902 (0.50)	0.004	0.155	0.090	0.594 (0.38)	1.223 (0.84)	0.465
Subcortical volumes	0.028 (0.08)	−0.041 (0.13)	0.640	−0.031	0.740	−0.054 (0.09)	0.340 (0.18)	0.045

NAS: Neuroimaging Association Scores; ADHD: Attention-Deficit/Hyperactivity Disorder. NAS shown in the table refer to Pt of 1. Values are represented as mean (SD). After Bonferroni correction, results were considered significant if they presented a $P < 0.017$ (0.05/3 outcomes). Bold values indicate significant results.

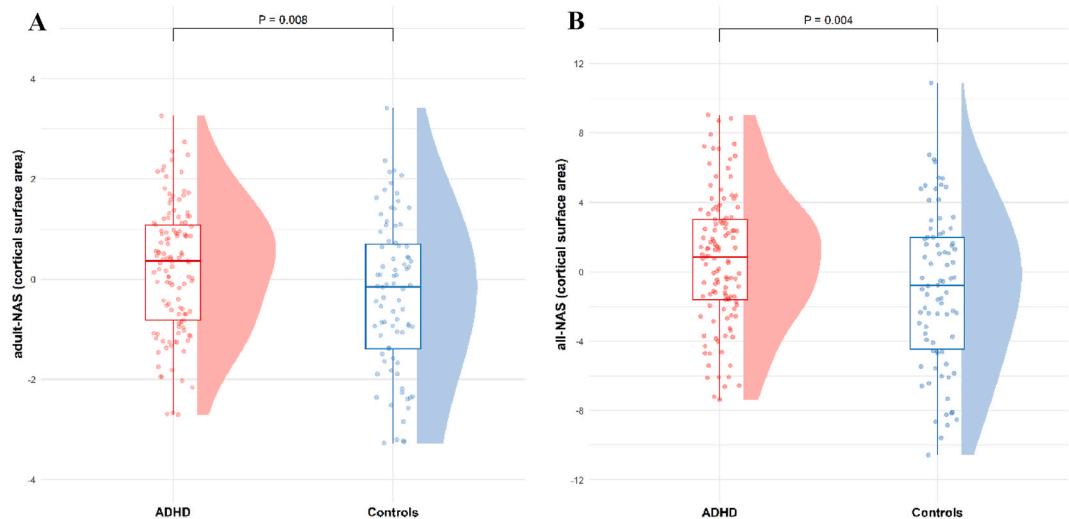


Fig. 1. Raincloud plot of adult- and all-NAS for cortical surface area. Results for the comparison between ADHD and controls for adult-NAS (A) and for all-NAS (B) are represented along with means and overall distribution of the scores.

methodology may hold promise to advance in the understanding of ADHD neurobiology.

Considering that brain differences are observed in children with ADHD and that symptoms often persist into adulthood, it is reasonable to expect atypical brain development accompanying symptom persistence into adulthood. Indeed, atypical patterns in ADHD during adolescence and adulthood have been identified, such as alterations in white matter microstructures (Chiang et al., 2023) and functional connectivity measured through EEG (Michellini et al., 2019). Our results extend the findings of Hoogman et al. (2019) to adulthood, highlighting the biological significance of cortical surface area in ADHD. Cortical surface area, a measure of gray matter, plays a crucial role in processing information and provides a biologically accurate depiction of the brain's cortical architecture (as demonstrated in Winkler et al. (2010)). Furthermore, these findings align with previous reports indicating that ADHD is one of the phenotypes with the highest negative genetic correlation with surface area (Grasby et al., 2020), a result replicated by Liu et al. (2022).

The significant associations for NAS in ADHD diagnostic status but not in its presentations, number of symptoms nor across midlife clinical trajectories suggest that, at this point, the technique was robust to capture a qualitative difference in the neurodevelopment of cases and controls but could still lack sufficient accuracy for more subtle aspects of clinical heterogeneity. This is in agreement with the original findings of Axelrud et al. (2021), where NAS predicted better general psychopathology than specific symptomatic domains.

For ADHD presentations, the lack of significant findings aligns with previous reviews indicating no differences between combined and inattentive presentations in most structural MRI studies (Saad et al.,

2020). Structural and functional connectivity differences, on the other hand, may better conceptualize ADHD presentations and explain its variations (Saad et al., 2020). Among studies reporting differences, some found smaller volumes in the combined presentation compared to inattentive, while others reported the opposite, but the brain regions implicated in each study differed (Anderson et al., 2014; Semrud-Clikeman et al., 2014; Al-Amin et al., 2018). These studies differ from ours in sample size and composition, with most associations found in predominantly male samples ranging from childhood to young adulthood.

Interestingly, we observed a trend towards an association between the clinical trajectories and all-NAS for subcortical volumes. Higher scores, indicating a greater similarity to an ADHD prototype brain, were found in subjects with an unstable trajectory. This is unexpected, as our hypothesis predicted higher scores in the stable trajectory group. Although previous studies did not report significant findings for subcortical volumes and ADHD trajectories (Proal et al., 2011; Luo et al., 2020), drawing conclusions is challenging due to heterogeneity in sample sizes, age ranges, methodologies, and sample compositions.

NAS for cortical surface area showed higher reliability indices compared to cortical thickness and subcortical volumes in children and adolescents (Axelrud et al., 2021). Similar findings were observed in our adult sample, indicating that adults followed the patterns reported for childhood ADHD. NAS also predicted general psychopathology with greater replicability than single structures alone, underscoring the method's validity (Axelrud et al., 2021). While these scores are in early stages and do not have clinical applications, they can be enhanced by increasing the power of discovery samples, particularly those groups underrepresented, such as adults. Similar to the development of polygenic scores, increasing sample size would improve the accuracy and

prediction power of the method (Demontis et al., 2023).

Our study should be viewed in light of some limitations. First, due to our sample size, we were not able to evaluate interaction effects regarding potential factors that may influence the relationship between brain aspects and ADHD, such as sex and comorbidities. Second, the acquisition of MRI data in only one time point (in the third wave of evaluation) does not allow us to investigate whether the brain's structural pattern of cortical surface area in ADHD remains constant or varies with symptom trajectories throughout midlife aging. Finally, the hyperactivity/impulsivity presentation was not evaluated due to its low prevalence. This limitation is usually seen in other studies as well.

5. Conclusion

Previous studies with larger sample sizes did not find structural brain differences between ADHD in adults and controls. In this study, we were able to detect such differences by improving statistical power through the combination of regional effects into a single score. This approach has been previously demonstrated to overcome the evaluation of each region separately. Moreover, the robustness of this measure has been reported in prior research on youth and our study has extended these findings to adults. Our results highlight the neurobiological relevance of the NAS measure and suggest that its application may enrich our understanding of ADHD in adulthood.

Role of the funding source

This study was financed by the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq, grants 466722/2014-1, 424041/2016-2, 426905/2016-2), Coordenação de Aperfeiçoamento de Pessoal de Nível Superior—Brasil (CAPES, Finance Code 001), FIPE-HCPA (16-0600, 21-0254), Fundação de Amparo à Pesquisa do Estado do Rio Grande do Sul (FAPERGS, grants 19/2551-0001731-6, 19/2551-0001668-9) and São Paulo Research Foundation (FAPESP, grant 2020/05652-0). The funding agencies mentioned were not involved in study design, in the collection, analysis and interpretation of data, in the writing of the report or in the decision to submit the article for publication.

CRediT authorship contribution statement

Cibele Edom Bandeira: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Eugenio Horacio Grevet:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Eduardo Schneider Vitola:** Writing – review & editing, Methodology. **Bruna Santos da Silva:** Writing – review & editing, Methodology, Conceptualization. **Renata Basso Cupertino:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Felipe Almeida Picon:** Writing – review & editing, Methodology. **Lucas Toshio Ito:** Writing – review & editing, Methodology. **Maria Eduarda de Araujo Tavares:** Writing – review & editing, Methodology. **Diego Luiz Rovaris:** Writing – review & editing, Resources. **Oliver Grimm:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Claiton Henrique Dotto Bau:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpsychires.2024.06.025>.

References

- Al-Amin, M., Zinchenko, A., Geyer, T., 2018. Hippocampal subfield volume changes in subtypes of attention deficit hyperactivity disorder. *Brain Res.* 1685, 1–8.
- Anderson, A., Douglas, P.K., Kerr, W.T., Haynes, V.S., Yuille, A.L., Xie, J., et al., 2014. Non-negative matrix factorization of multimodal MRI, fMRI and phenotypic data reveals differential changes in default mode subnetworks in ADHD. *Neuroimage* 102 (Pt 1), 207–219.
- Axelrud, L.K., Simioni, A.R., Pine, D.S., Winkler, A.M., Pan, P.M., Sato, J.R., et al., 2021. Neuroimaging Association Scores: reliability and validity of aggregate measures of brain structural features linked to mental disorders in youth. *Eur. Child Adolesc. Psychiatr.* 30 (12), 1895–1906.
- Bandeira, C.E., Grevet, E.H., Cupertino, R.B., Tavares, M.E. de A., Gusmão, C.S., Kappel, D.B., et al., 2021. The role of glucocorticoid receptor gene in the association between attention deficit-hyperactivity disorder and smaller brain structures. *J. Neural. Transm.* 128 (12), 1907–1916.
- Chiang, H.-L., Tseng, W.-Y.I., Tseng, W.-L., Tung, Y.-H., Hsu, Y.-C., Chen, C.-L., et al., 2023. Atypical development in white matter microstructures in ADHD: a longitudinal diffusion imaging study. *Asian J Psychiatr* 79, 103358.
- Demontis, D., Walters, G.B., Athanasiadis, G., Walters, R., Therrien, K., Nielsen, T.T., et al., 2023. Genome-wide analyses of ADHD identify 27 risk loci, refine the genetic architecture and implicate several cognitive domains. *Nat. Genet.* 55 (2), 198–208.
- Faraone, S.V., Biederman, J., Mick, E., 2006. The age-dependent decline of attention deficit hyperactivity disorder: a meta-analysis of follow-up studies. *Psychol. Med.* 36 (2), 159–165.
- Fireouabadi, F.D., Ramezanpour, S., Fireouabadi, M.D., Yousem, I.J., Puts, N.A.J., Yousem, D.M., 2022. Neuroimaging in attention-deficit/hyperactivity disorder: recent advances. *AJR Am. J. Roentgenol.* 218 (2), 321–332.
- First, M.B., Spitzer, R.L., Gibbon, M., Williams, J.B.W., 1998. Structured Clinical Interview for DSM-IV Axis I Disorders, Research Version, Patient Edition (SCID-I/P), first ed. Amer Psychiatric Pub Inc, p. 132.
- Fischl, B., 2012. FreeSurfer. *Neuroimage* 62 (2), 774–781.
- George, D., Mallery, P., 2003. Hints and Answers to Selected Exercises. SPSS for Windows Step by Step: A Simple Guide and Reference 110 Update Wps Ablongman Com/wps/media/objects/385 George 4answers Pdf.
- Gilliam, M., Stockman, M., Malek, M., Sharp, W., Greenstein, D., Lalonde, F., et al., 2011. Developmental trajectories of the corpus callosum in attention-deficit/hyperactivity disorder. *Biol. Psychiatr.* 69 (9), 839–846.
- Grasby, K.L., Jahanshad, N., Painter, J.N., Colodro-Conde, L., Bralten, J., Hibar, D.P., et al., 2020. The genetic architecture of the human cerebral cortex. *Science* 367 (6484).
- Grevet, E.H., Bandeira, C.E., Vitola, E.S., de Araujo Tavares, M.E., Breda, V., Zeni, G., et al., 2024. The course of attention-deficit/hyperactivity disorder through midlife. *Eur. Arch. Psychiatr. Clin. Neurosci.* 274 (1), 59–70.
- Grevet, E.H., Bau, C.H.D., Salgado, C.A.L., Ficher, A., Victor, M.M., Garcia, C., et al., 2005. [Interrater reliability for diagnosis in adults of attention deficit hyperactivity disorder and oppositional defiant disorder using K-SADS-E]. *Arq Neuropsiquiatr* 63 (2A), 307–310.
- Hibar, D.P., Stein, J.L., Renteria, M.E., Arias-Vasquez, A., Desrivieres, S., Jahanshad, N., et al., 2015. Common genetic variants influence human subcortical brain structures. *Nature* 520 (7546), 224–229.
- Hoogman, M., Bralten, J., Hibar, D.P., Mennes, M., Zwiers, M.P., Schwenen, L.S.J., et al., 2017. Subcortical brain volume differences in participants with attention deficit hyperactivity disorder in children and adults: a cross-sectional mega-analysis. *Lancet Psychiatr.* 4 (4), 310–319.
- Hoogman, M., Muetzel, R., Guimaraes, J.P., Shumskaya, E., Mennes, M., Zwiers, M.P., et al., 2019. Brain imaging of the cortex in ADHD: a coordinated analysis of large-scale clinical and population-based samples. *Am. J. Psychiatr.* 176 (7), 531–542.
- Klein, M., Souza-Duran, F.L., Menezes, A.K.P.M., Alves, T.M., Busatto, G., Louzã, M.R., 2021. Gray matter volume in elderly adults with ADHD: associations of symptoms and comorbidities with brain structures. *J. Atten. Disord.* 25 (6), 829–838.
- Kochunov, P., Huang, J., Chen, S., Li, Y., Tan, S., Fan, F., et al., 2019. White matter in schizophrenia treatment resistance. *Am. J. Psychiatr.* 176 (10), 829–838.
- Lancaster, T.M., Dimitriadis, S.I., Perry, G., Zammit, S., O'Donovan, M.C., Linden, D.E., 2022. Morphometric analysis of structural MRI using schizophrenia meta-analytic priors distinguish patients from controls in two independent samples and in a sample of individuals with high polygenic risk. *Schizophr. Bull.* 48 (2), 524–532.
- Liu, S., Smit, D.J.A., Abdellaoui, A., van Wingen, G.A., Verweij, K.J.H., 2022. Brain structure and function show distinct relations with genetic predispositions to mental health and cognition. *Biol Psychiatry Cogn Neurosci Neuroimaging* 8 (3), 300–310.
- Luo, Y., Halperin, J.M., Li, X., 2020. Anatomical substrates of symptom remission and persistence in young adults with childhood attention deficit/hyperactivity disorder. *Eur. Neuropsychopharmacol* 33, 117–125.
- Mercadante, M.T., Asbarh, F., Rosário, M.C., Ayres, A.M., Ferrari, M.C., Assumpção, F.B., et al., 1995. K-SADS, entrevista semi-estruturada para diagnóstico em psiquiatria da infância, versão epidemiológica. 1ª ed. São Paulo: PROTOC Hospital das Clínicas da FMUSP.
- Micheline, G., Jurgiel, J., Bakolis, I., Cheung, C.H.M., Asherson, P., Loo, S.K., et al., 2019. Atypical functional connectivity in adolescents and adults with persistent and remitted ADHD during a cognitive control task. *Transl. Psychiatry* 9 (1), 137.
- Oldfield, R.C., 1971. The assessment and analysis of handedness: the Edinburgh inventory. *Neuropsychologia* 9 (1), 97–113.
- Proal, E., Reiss, P.T., Klein, R.G., Mannuzza, S., Gotimer, K., Ramos-Olzagasti, M.A., et al., 2011. Brain gray matter deficits at 33-year follow-up in adults with attention-deficit/hyperactivity disorder established in childhood. *Arch. Gen. Psychiatr.* 68 (11), 1122–1134.

- Revelle, W., 2017. Psych: Procedures for Personality and Psychological Research.
- Rodrigue, A.L., Hayes, R.A., Waite, E., Corcoran, M., Glahn, D.C., Jalbrzikowski, M., 2023. Multimodal neuroimaging summary scores as neurobiological markers of psychosis. *Schizophr. Bull.* 2023 Oct 16:sbad149. Epub ahead of print.
- Saad, J.F., Griffiths, K.R., Korgaonkar, M.S., 2020. A systematic review of imaging studies in the combined and inattentive subtypes of attention deficit hyperactivity disorder. *Front. Integr. Neurosci.* 14, 31.
- Semrud-Clikeman, M., Pliszka, S.R., Bledsoe, J., Lancaster, J., 2014. Volumetric MRI differences in treatment naïve and chronically treated adolescents with ADHD-combined type. *J. Atten. Disord.* 18 (6), 511–520.
- Shaw, P., Eckstrand, K., Sharp, W., Blumenthal, J., Lerch, J.P., Greenstein, D., et al., 2007. Attention-deficit/hyperactivity disorder is characterized by a delay in cortical maturation. *Proc. Natl. Acad. Sci. U.S.A.* 104 (49), 19649–19654.
- Shaw, P., Malek, M., Watson, B., Sharp, W., Evans, A., Greenstein, D., 2012. Development of cortical surface area and gyrification in attention-deficit/hyperactivity disorder. *Biol. Psychiatr.* 72 (3), 191–197.
- Wang, Y., Zuo, C., Xu, Q., Hao, L., Zhang, Y., 2021. Attention-deficit/hyperactivity disorder is characterized by a delay in subcortical maturation. *Prog. Neuro-Psychopharmacol. Biol. Psychiatry* 104, 110044.
- Winkler, A.M., Kochunov, P., Blangero, J., Almasy, L., Zilles, K., Fox, P.T., et al., 2010. Cortical thickness or grey matter volume? The importance of selecting the phenotype for imaging genetics studies. *Neuroimage* 53 (3), 1135–1146.
- Zhang-James, Y., Helminen, E.C., Liu, J., Enigma-Adhd Working Group, Franke, B., Hoogman, M., et al., 2021. Evidence for similar structural brain anomalies in youth and adult attention-deficit/hyperactivity disorder: a machine learning analysis. *Transl. Psychiatry* 11 (1), 82.